Practice Questions

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Score: 82% Passing Score: 80%

Question 1: ✔ Correct

You need to transfer data from one laptop to another, and you would like to use an Ethernet cable. You do not have a hub or a switch.

Which type of cable should you use?

- Straight-through
- ➡ Crossover
- Loopback
- Rollover

EXPLANATION

Use a crossover cable to connect two devices together in a back-to-back configuration.

Use a straight-through cable to connect a workstation to a hub or switch port. Use a rollover cable to connect a workstation to the console port of a router or a switch. Use a loopback plug to allow a device to communicate with itself through its own network adapter.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 2: ✔ Correct

You want to connect the LAN port on a router to the uplink port on a switch. The switch does not support auto-MDI.

Which type of cable should you use?

- Loopback
- ➡ Crossover
- Straight-through
- Rollover

EXPLANATION

Use a crossover cable to connect a workstation or a router to the uplink port on a switch.

Use a straight-through cable to connect the router to a regular switch port. Use a rollover cable to connect a workstation to the console port of a router. Use a loopback plug to allow a device to communicate with itself through its own network adapter.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 3: ✔ Correct

You need to connect two switches using their uplink ports. The switches do not support auto-MDI.

Which type of cable should you use?

- Rollover
- ➡ Crossover
- Loopback
- Straight-through

EXPLANATION

Use a crossover cable to connect two switches through their uplink ports or to connect two switches through regular ports.

Use a straight-through cable to connect the uplink port on one switch to a regular port on another switch. Use a rollover cable to connect a workstation to the console port of the switch. Use a loopback plug connected to a single port for troubleshooting.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 4: ✕ Incorrect

You need to connect a workstation to a switch using a regular port on the switch (not an uplink port). The switch does not support auto-MDI.

Which type of cable should you use?

- ➡ Straight-through
- ~~Crossover~~
- Loopback
- Rollover

EXPLANATION

Use a straight-through cable to connect a workstation or router to a regular switch port.

Use a crossover cable to connect the workstation to the uplink port. Use a rollover cable to connect the workstation to the console port of the switch. Use a loopback plug to allow a workstation to communicate with itself through its own network adapter.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 5: ✔ Correct

You have purchased a new router that you need to configure. You need to connect a workstation to the router's console port to complete the configuration tasks.

Which type of cable would you most likely use?

- RG6
- Straight-through
- Crossover
- ➡ Rollover

EXPLANATION

Use a rollover cable to connect a workstation to the console port of a router or a switch. The rollover cable has an RJ45 connector on one end to connect to the console port. The other end has a serial connector to connect to the serial port of the workstation. You then run a terminal emulation program on the workstation to connect to the console of the router or switch to perform configuration and management tasks.

Use a straight-through or crossover Ethernet cable to connect devices using the Ethernet RJ45 ports. An RG6 cable is a coaxial cable.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 6: ✔ Correct

Which of the following connectors is typically used on one end of a rollover cable?

- BNC
- SC
- LC
- ➡ Serial
- F-type
- ST

EXPLANATION

A rollover cable has a serial connector on one end and an RJ45 connector on the other end. Alternatively, it might have an RJ45 connector on both ends and use a serial converter to convert from the RJ45 connector to a serial connector.

BNC and F-type connectors are used with coaxial cables. ST, SC, and LC connectors are used with fiber optic cables.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 7: ✔ Correct

Which of the following standards is typically used in a rollover cable?

- RJ11
- RG58
- ➡ RS232
- RG6

EXPLANATION

A rollover cable has a serial connector on one end and an RJ45 connector on the other end. RS232 is the standard for serial communications.

RJ11 connectors are used for analog telephone lines. RG6 and RG58 are coaxial cable standards.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 8: ✔ Correct

This question includes an image to help you answer the question. View Image

Use the exhibit to match the connector type on the left with the corresponding letter on the right.

A

✔ DB25

B

✔ DB9

C

✔ RJ45

D

✔ RJ11

E

✔ LC

F

✔ BNC

G

✔ F-Type

EXPLANATION

- **A** DB25 connectors are older serial connectors.
- **B** DB9 connectors are usually on the ends of RS232 serial cables.
- **C** RJ45 connectors are used for Ethernet networking with twisted pair cables.
- **D** RJ11 connectors are used for dial-up and some DSL internet connections.
- **E** LC connectors are used with fiber optic cables.
- **F** BNC connectors are used with coaxial cables on 10Base2 Ethernet networks.
- **G** F-Type connectors are used with coaxial cables that create cable TV and broadband cable connections.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 9: ✕ Incorrect

You are implementing a SOHO network for a local business. The ISP has already installed and connected a cable modem in the business.

The business has four computers that need to communicate with each other and the internet. The ISP's cable modem has only one RJ45 port. You need to set up the network within the following parameters:

- You must spend as little money as possible.
- You must not purchase unnecessary equipment.
- Computers need to have a gigabit connection to the network.
- New devices should not require management or configuration.

You examine each computer and notice only one of the four computers has a wireless NIC; they all have Ethernet NICs.

What should you purchase?

- ➡ An unmanaged switch and CAT 5e cabling.
- A hub and CAT 5e cabling.
- ~~A wireless AP and three new wireless NICs.~~
- A managed switch and CAT 6 cabling.
- A new cable modem with a built-in switch and CAT 6a cabling.

EXPLANATION

You should purchase an unmanaged switch and CAT 5e cabling. Switches offer guaranteed bandwidth to each switch port and full-duplex communication. Unmanaged switches are autonomous in their function, requiring no port management or configuration. CAT 5e cabling supports transfer speeds up to 1000 Mbps.

Purchasing a new cable modem with a built-in switch would be more expensive than an unmanaged switch. Additionally, CAT6a cabling is unnecessary for this type of network. Because all of the computers already have wired NICs, purchasing a wireless AP and three new wireless NICs would introduce new costs. The wireless AP would also require additional management and configuration. Hubs suffer from collisions, so only half-duplex communication is possible. This wouldn't support 1000 Mbps speeds. A small business with four computers doesn't need the additional features that a managed switch provides.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 10: ✔ Correct

This question includes an image to help you answer the question. View Image

You manage the two-location network shown in the exhibit. Workstations and servers at each location connect to a patch panel using behind-the-wall wiring. The patch panel then connects network hosts to one of three 1000BASE-T switches. Routers are implemented at each location to connect the two networks together using a private WAN. The switch ports have auto-MDIX disabled.

Drag the cable type on the left to the most appropriate network location on the right. Each cable type can be used more than once.

A. Drop cables

✔ Cat 6 straight-through UTP

B. Patch cables

✔ Cat 6 straight-through UTP

C. Uplink cables

✔ Cat 6 crossover UTP

D. WAN cables

✔ Single-mode fiber optic

EXPLANATION

In this scenario, the following cables should be used in the following locations:

- Drop cables = Cat 6 straight-through UTP.
- Patch cables = Cat 6 straight-through UTP.
- Uplink cables = Cat 6 crossover UTP (Cat 6 straight-through UTP can be used instead if auto-MDIX is enabled on the switch ports).
- WAN cable = Single-mode fiber optic.

Because 1000BASE-T Ethernet switches are in use at both locations, Cat 5 or better UTP must be used. Because the distance between locations is greater than 550 meters, single-mode fiber optic cable must be used. This will require the use of 1000BASE-LX or 1000BASE-EX transceivers on both routers.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 11: ✔ Correct

You want to create a rollover cable that has an RJ45 connector on both ends. How should you connect the wires within the connectors?

- Connect pin 1 to pin 3 and pin 2 to pin 6.
- Connect each pin on one end to the same pin on the other end (pin 1 to pin 1, pin 2 to pin 2, etc.).
- Use the T568A standard on one end and the T568B standard on the other end.
- ➡ Connect pin 1 to pin 8, pin 2 to pin 7, pin 3 to pin 6, and pin 4 to pin 5.

EXPLANATION

When terminated with an RJ45 connector on both ends, the wires within the connectors are rolled over to the opposite connector as follows:

- Pin 1 connects to pin 8
- Pin 2 connects to pin 7
- Pin 3 connects to pin 6
- Pin 4 connects to pin 5

A crossover cable uses the T568A standard on one end and the T568B standard on the other end. The crossover cable connects pin 1 to pin 3 and pin 2 to pin 6. Connecting each pin to the same pin on the other end creates a straight-through cable.

REFERENCES

LabSim for Network Pro, Section 4.3.

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